Online POMDP Methods

Approximate POMDP Solutions

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

↓

Offline

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

Offline

Online

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

Offline

Previously

Online

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

Formulation Approximations

(solve a slightly different problem)

Offline

Previously

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Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

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Previously

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Formulation Approximations

(solve a slightly different problem)

Last Time

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

Offline

Previously

Online

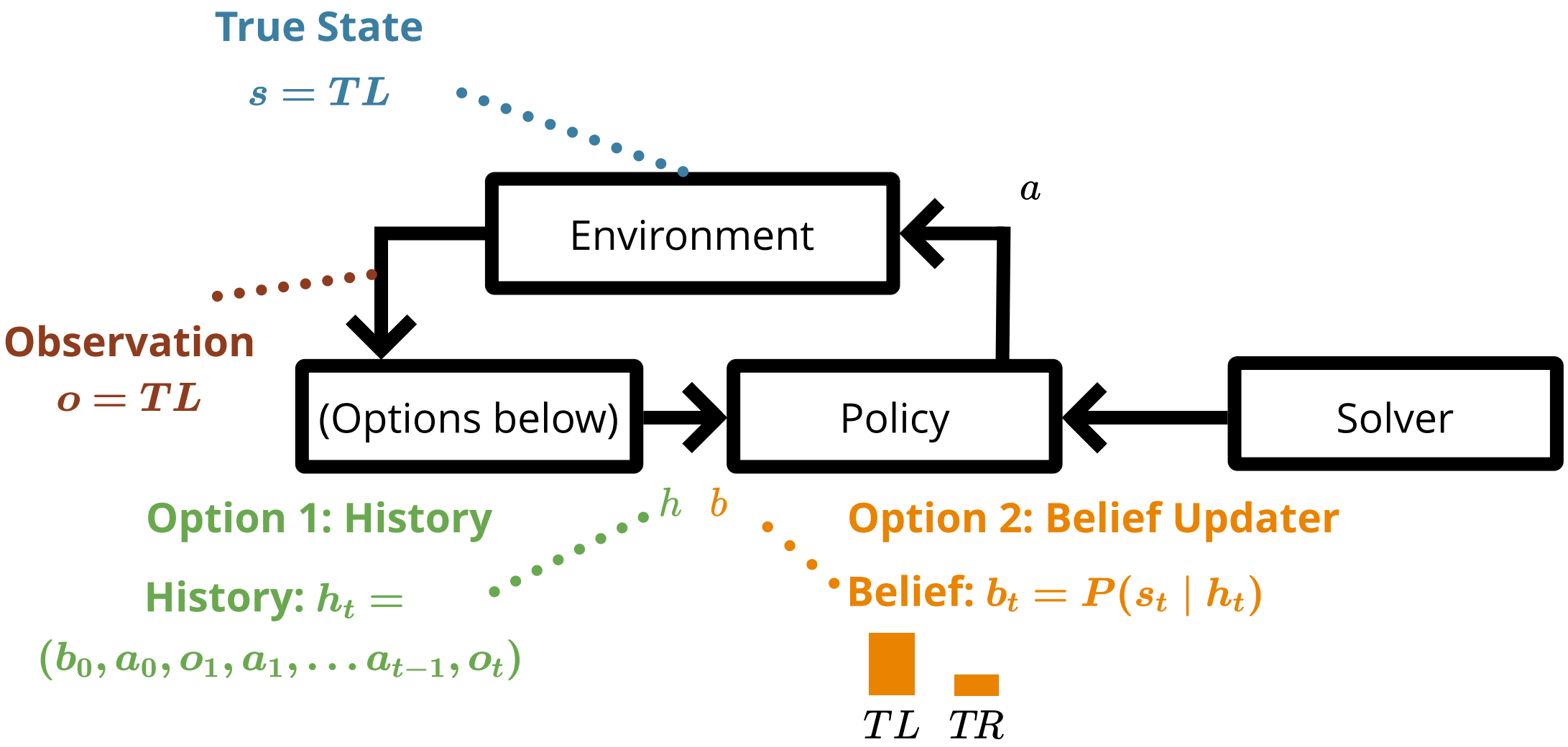
Today!

Formulation Approximations

(solve a slightly different problem)

Last Time

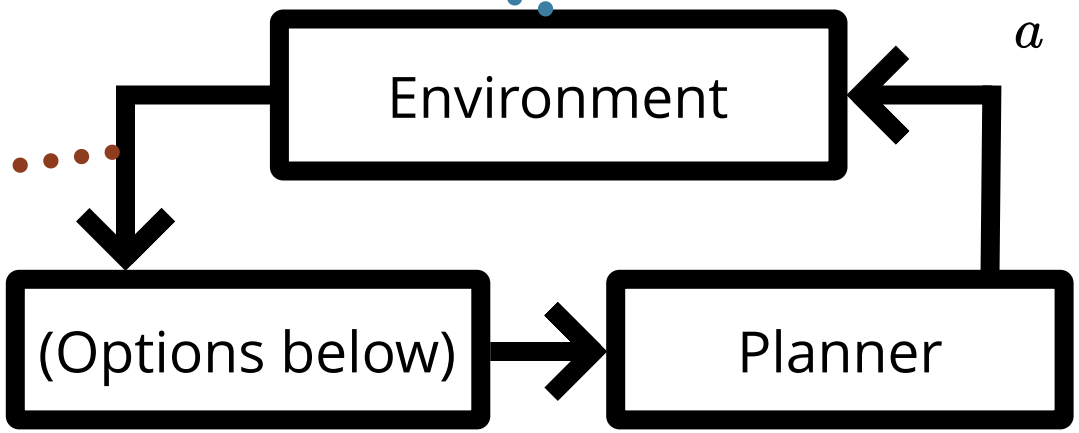
POMDP Sense-Plan-Act Loop



POMDP Sense-Plan-Act Loop

True State

$s = TL$



Observation

$o = TL$

Option 1: History

h

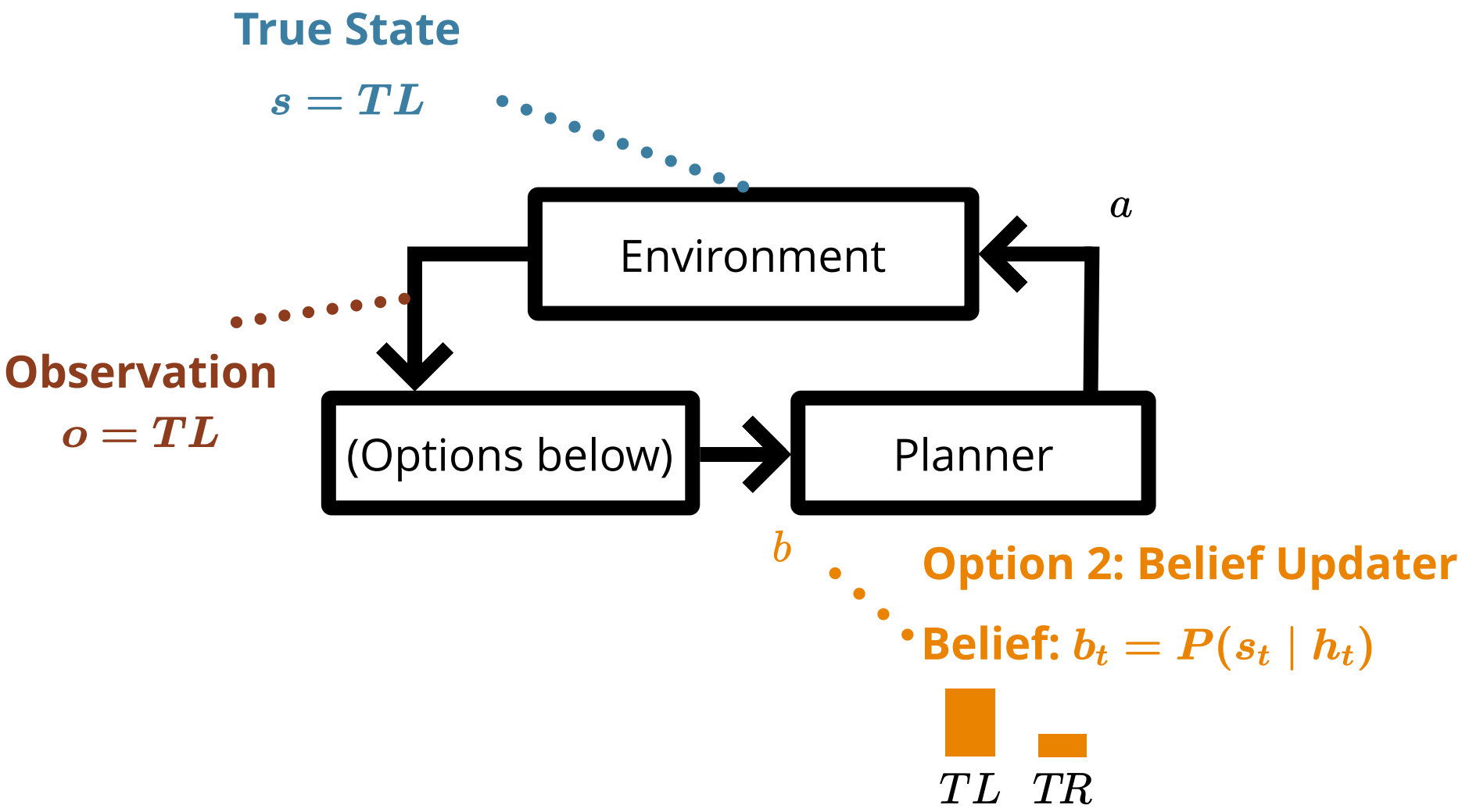
History: $h_t = (b_0, a_0, o_1, a_1, \ldots a_{t-1}, o_t)$

b

Option 2: Belief Updater

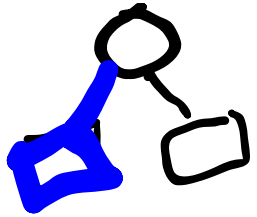
Belief: $b_t = P(s_t \mid h_t)$

POMDP Sense-Plan-Act Loop

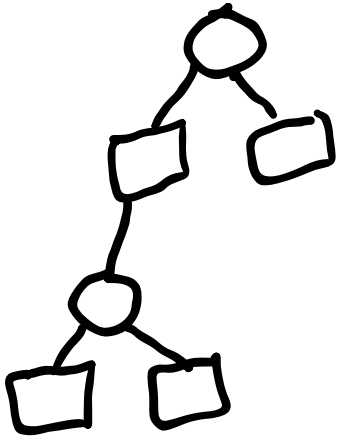


Monte Carlo Tree Search (MCTS/UCT)

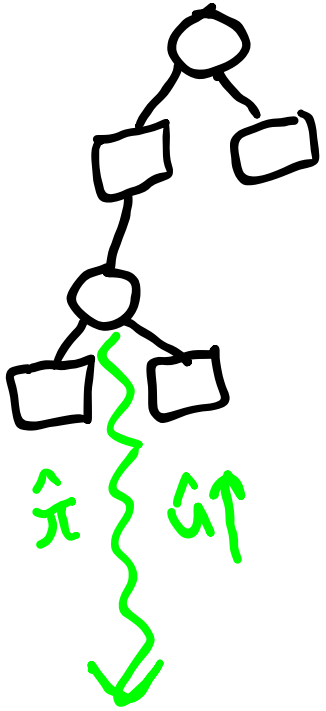
Search



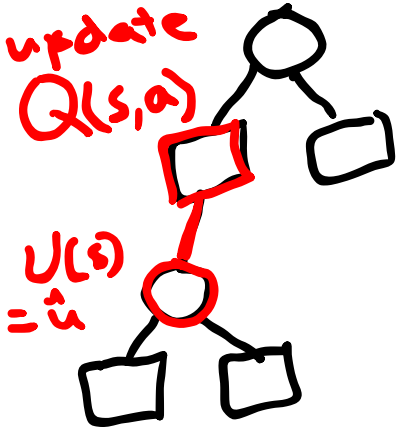
Expansion



Rollout



Backup



$$Q(s,a) + c\sqrt{\frac{\log N(s)}{N(s,a)}}$$

low $N(s,a)/N(s)$ = high bonus

start with $c = 2(\bar{V} - \underline{V})$

How should we adapt MCTS for POMDPs?

True State

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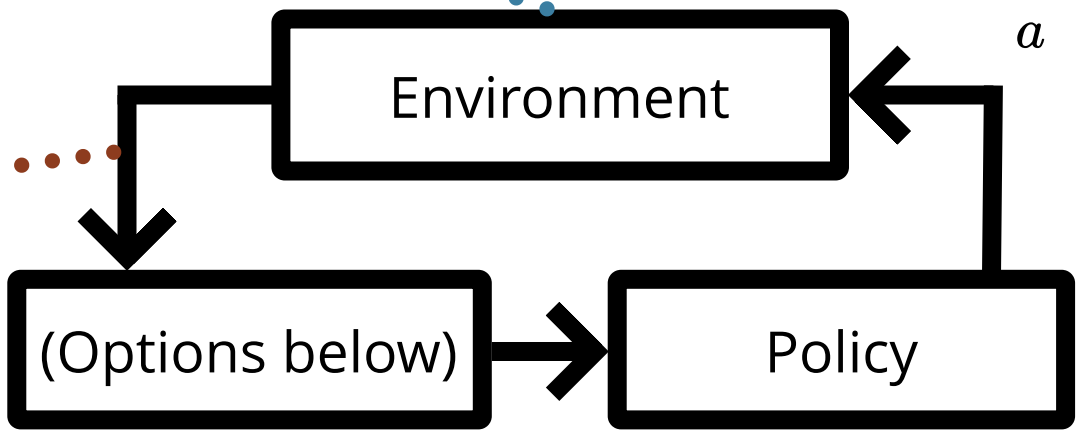

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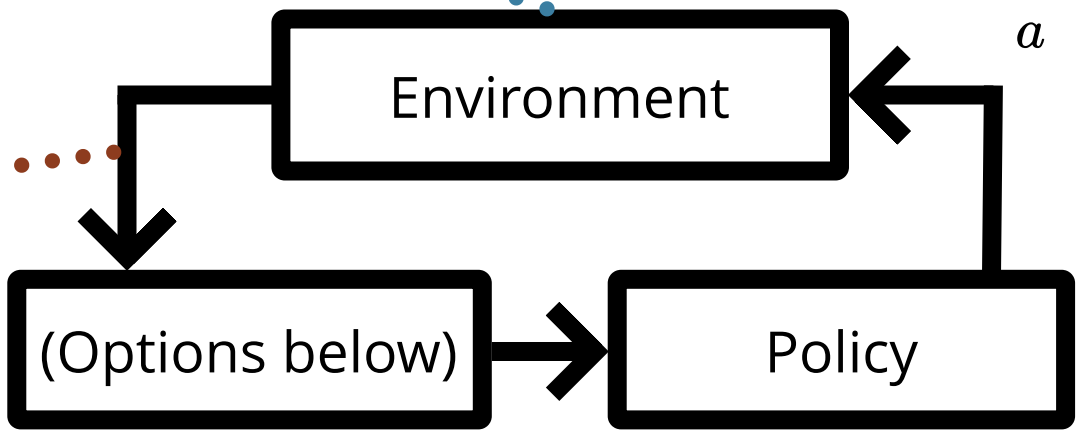

Observation

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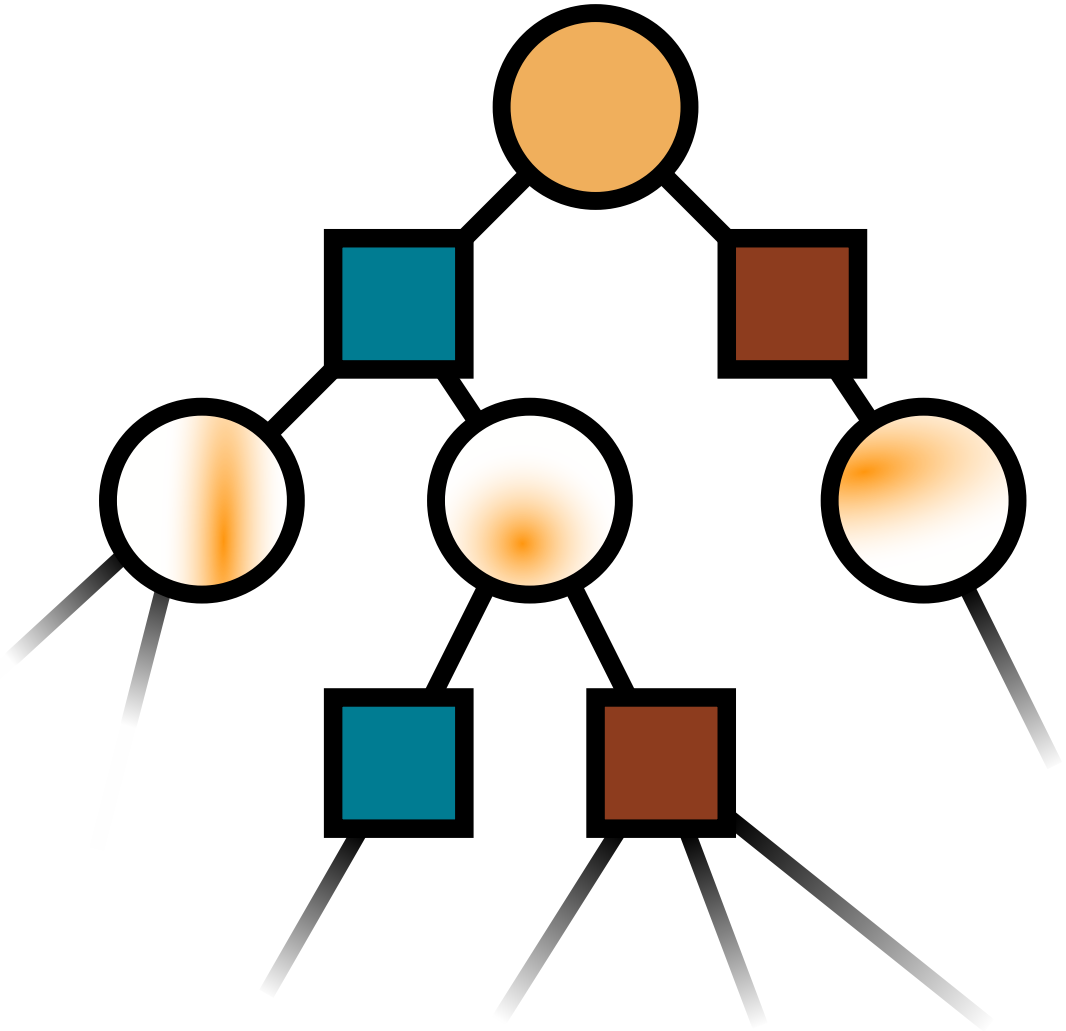
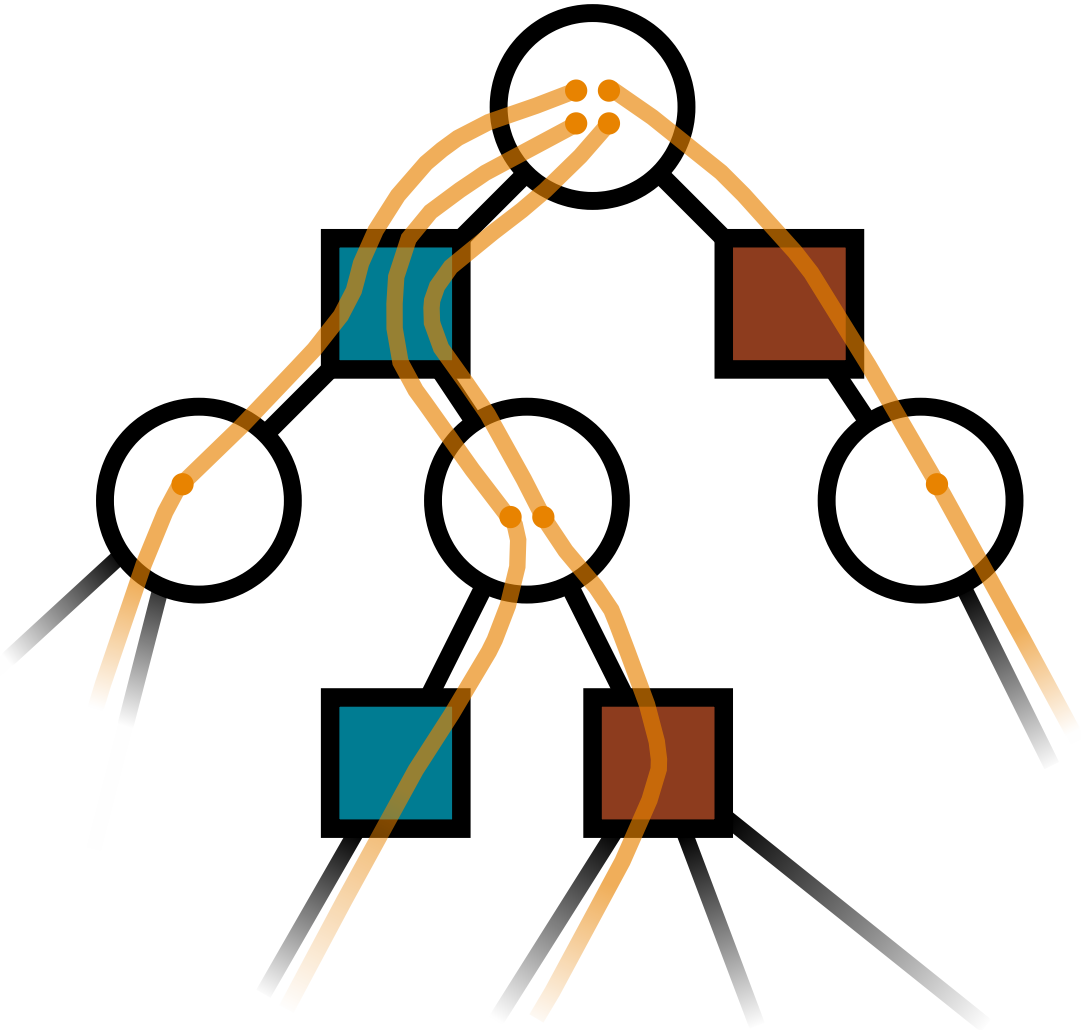
$(b_0, a_0, o_1, a_1, \ldots a_{t-1}, o_t)$

MCTS on Histories

Algorithm 6.12 Monte Carlo tree search

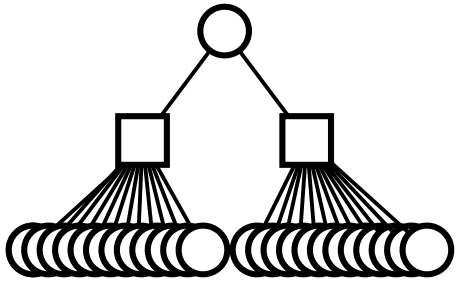
```
function SelectAction(b, d)
    h ← ∅
    loop
        s ~ b
        Simulate(s, h, d)
    return arg max_a Q(h, a)
function Simulate(s, h, d)
    if d = 0
        return 0
    if h ∉ T
        for a ∈ A(s)
            (N(h, a), Q(h, a)) ← (N_0(h, a), Q_0(h, a))
        T = T ∪ {h}
        return Rollout(s, d, π_0)
    a ← arg max_a [ Q(h, a) + c √( log N(h) / N(h, a) ) ]
    (s′, o, r) ~ G(s, a)
    q ← r + γ Simulate(s′, hao, d − 1)
    N(h, a) ← N(h, a) + 1
    Q(h, a) ← Q(h, a) + (q − Q(h, a)) / N(h, a)
    return q
```

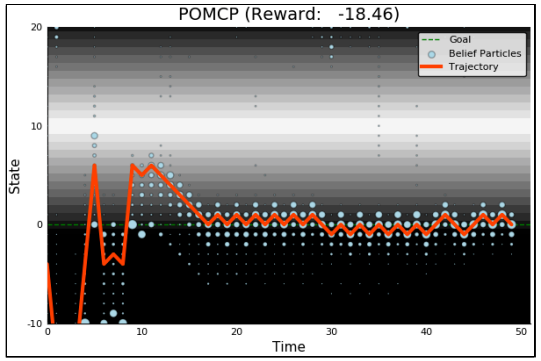




Continuous Observation Spaces

POMCP





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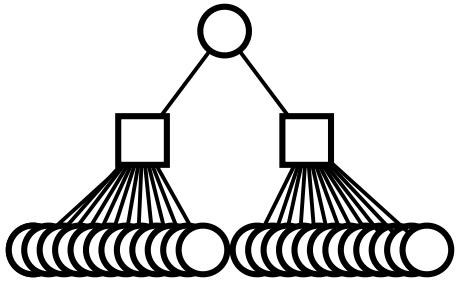

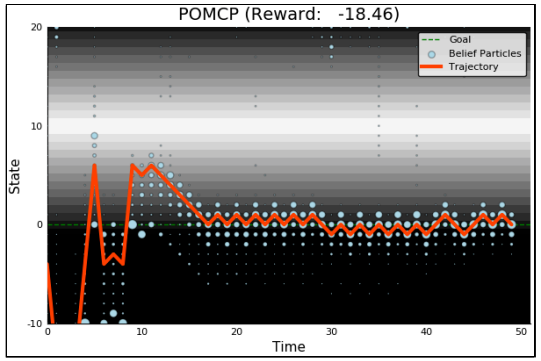

POMCPOW

Continuous Observation Spaces

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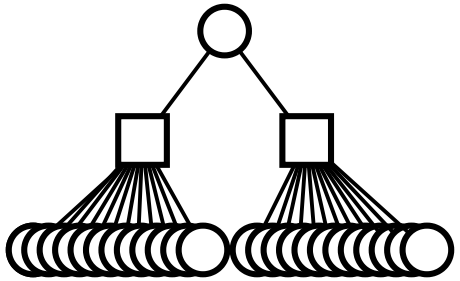

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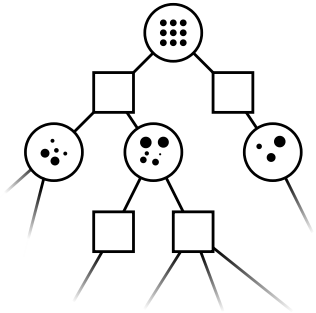

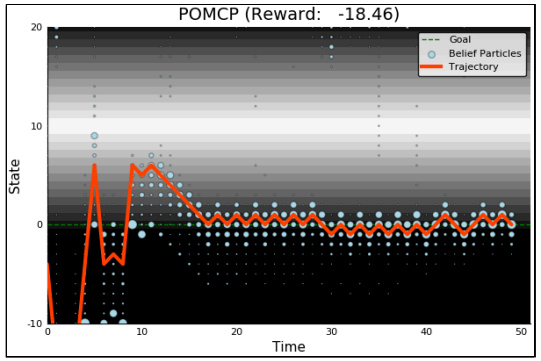

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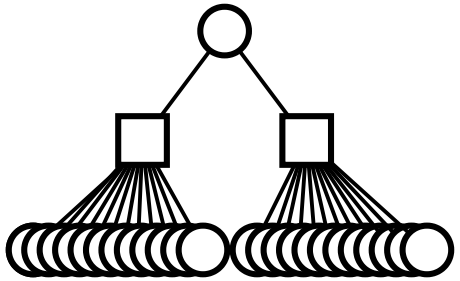

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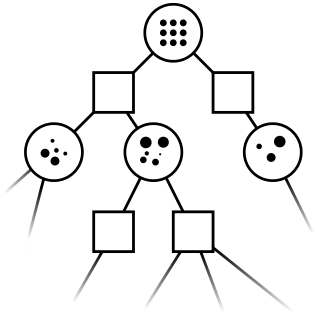

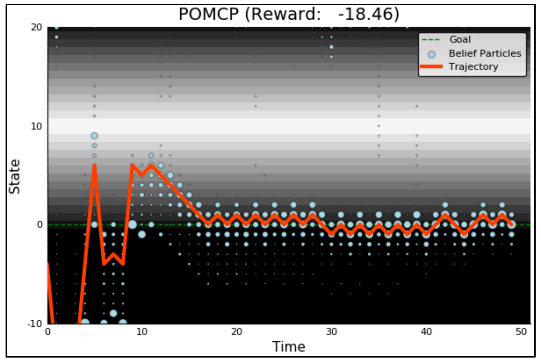

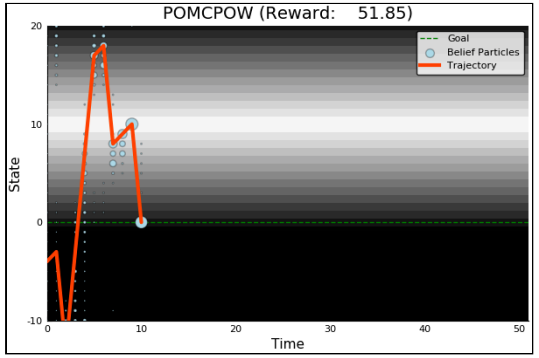

PF Approximation Accuracy

[Lim, Becker, Kochenderfer, Tomlin, & Sunberg, 2023]

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$\mathbf{M_P}$ = Particle belief MDP approximation of POMDP $\mathbf{P}$

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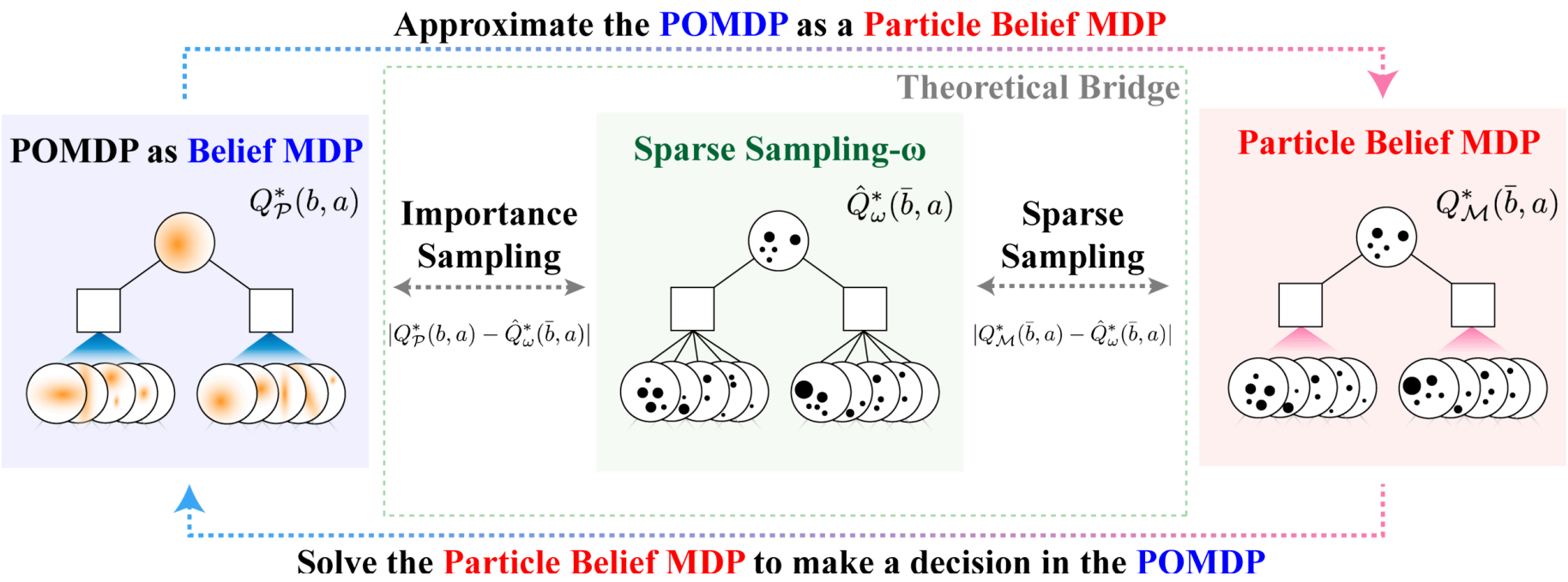

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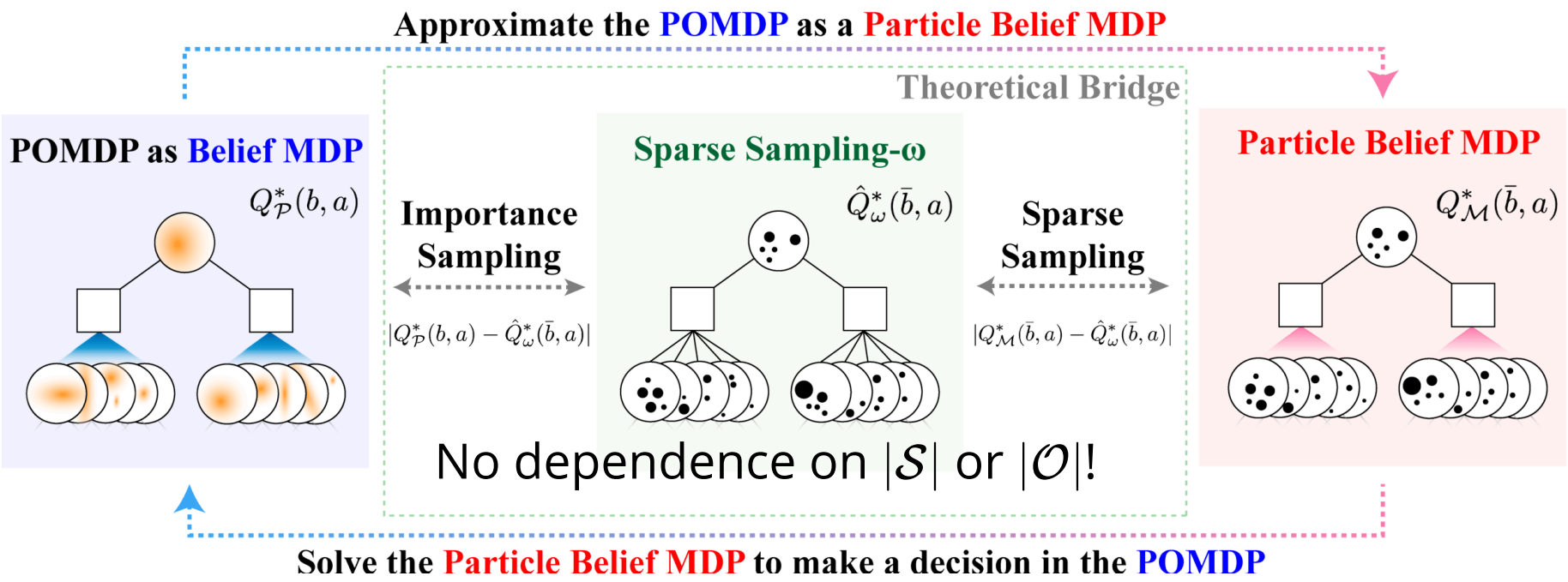

[Lim, Becker, Kochenderfer, Tomlin, & Sunberg, 2023]

Two Caveats

[Lee, Wray, Sunberg, Ahmed, ICRA 2026]

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Possible Solution to 2:

Rao-Blackwellized POMDP Planning

[Lee, Wray, Sunberg, Ahmed, ICRA 2026]

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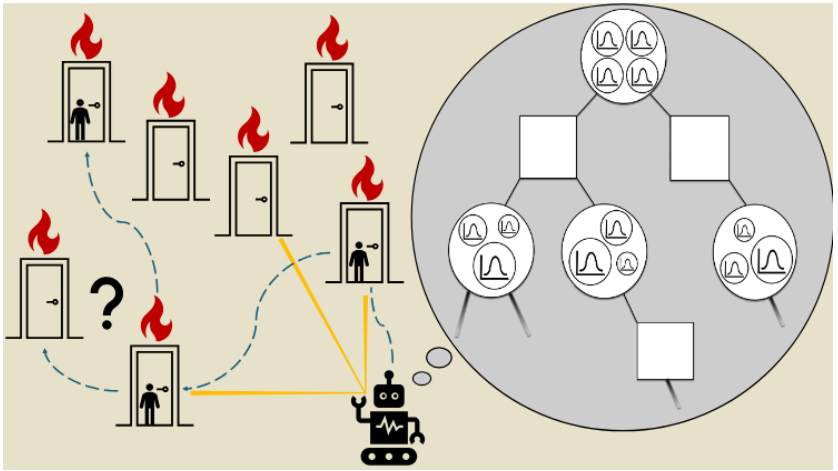

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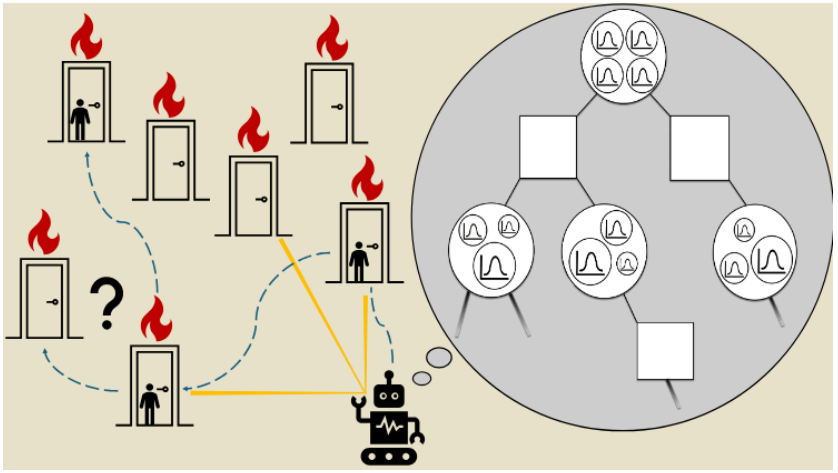

$$p(s_t \mid h_t)$$

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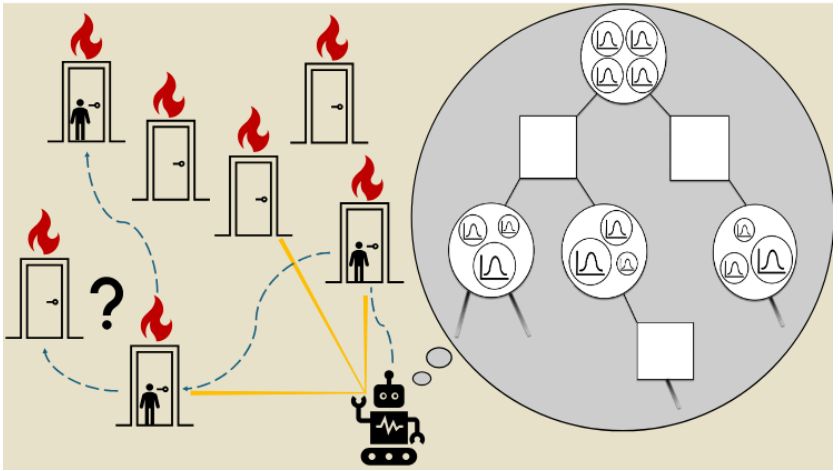

$$p(s_t \mid h_t)$$
$$= p(s_t^a \mid s_t^p, h_t)\, p(s_t^p \mid h_t)$$

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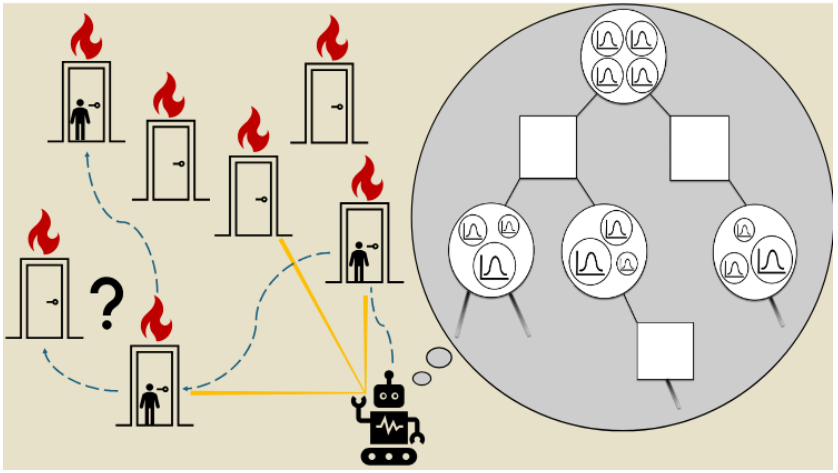

$$p(s_t \mid h_t)$$
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Analytical update (e.g. Kalman Filter)

[Lee, Wray, Sunberg, Ahmed, ICRA 2026]

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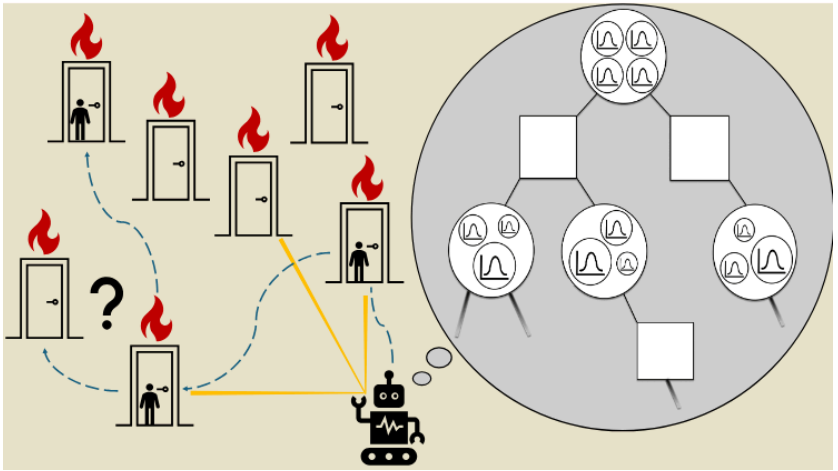

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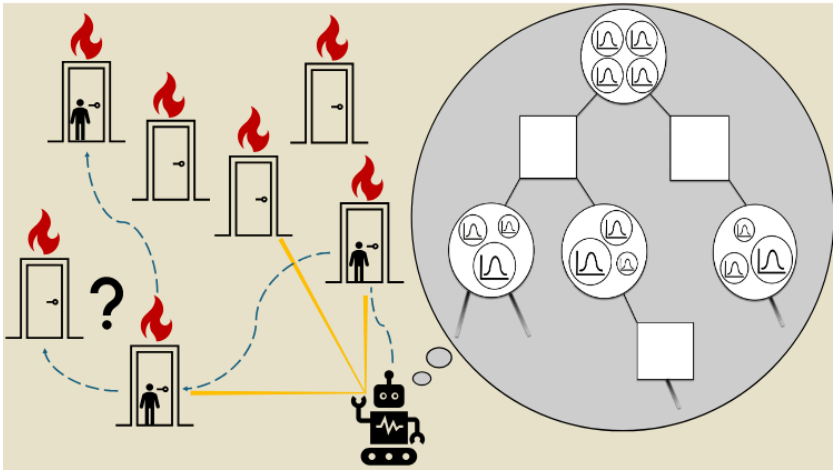

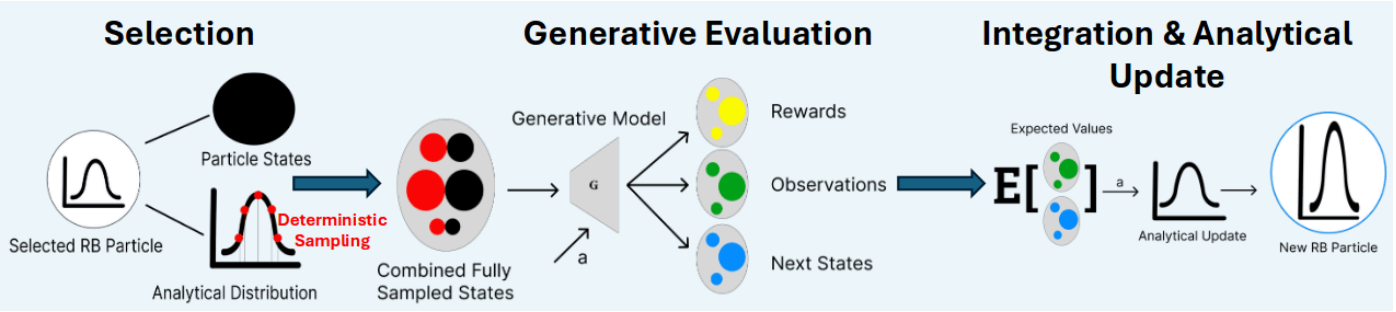

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Continuous Action Spaces

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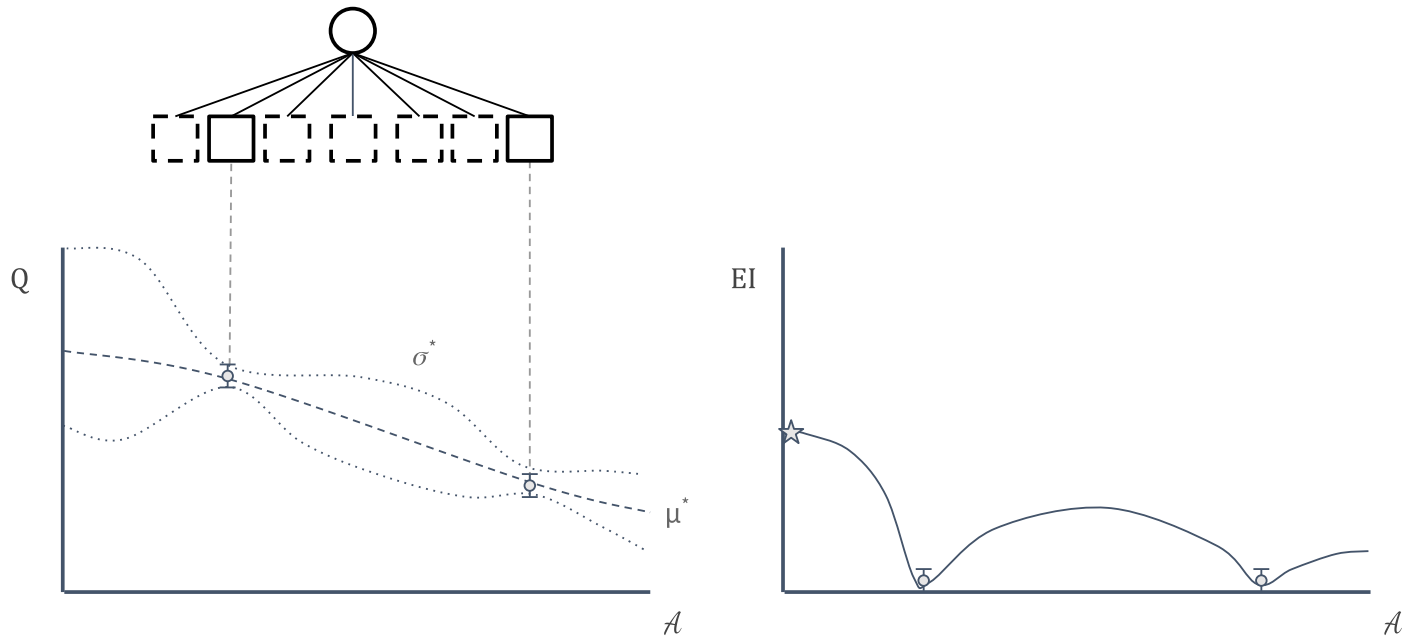
BOMCP

Bayesian Optimized Action Branching



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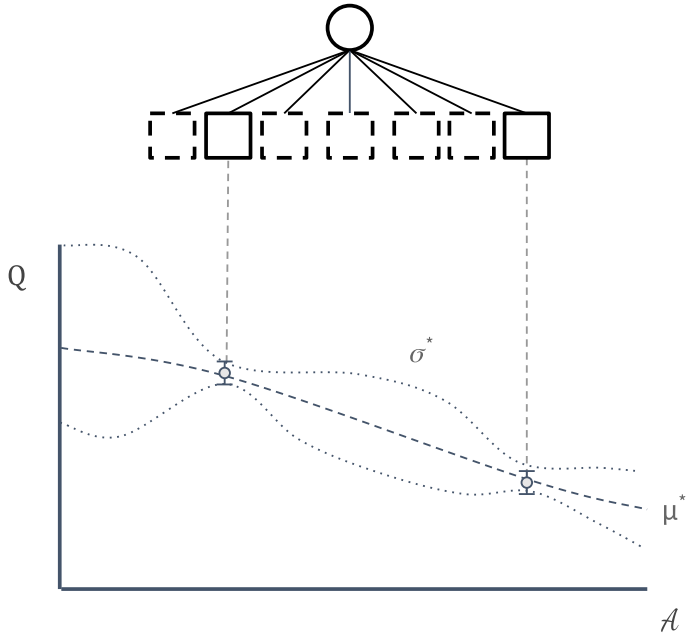

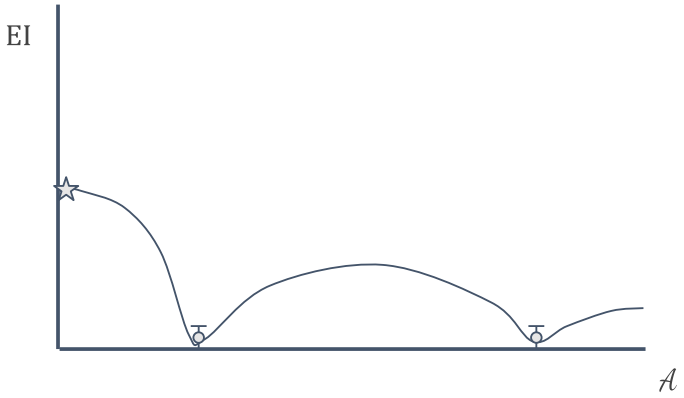

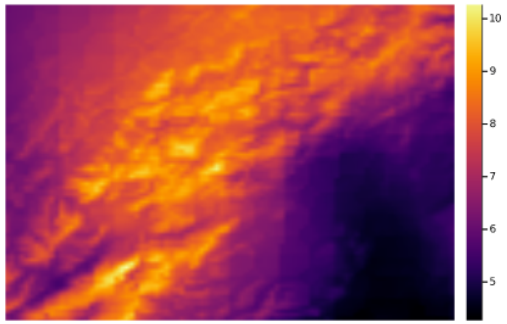

Figure 2: Wind Map. Figure shows wind map for Altamont Pass, CA at 100m altitude. The colors represent the average annual wind speed in m/s.

Continuous Action Spaces

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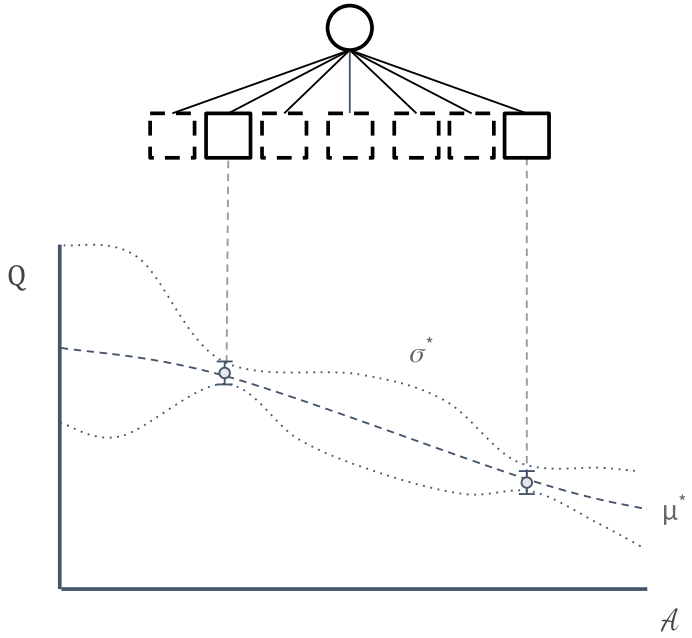
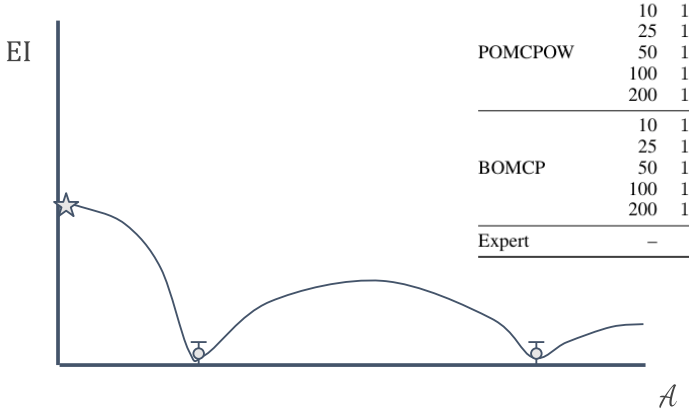
Bayesian Optimized Action Branching





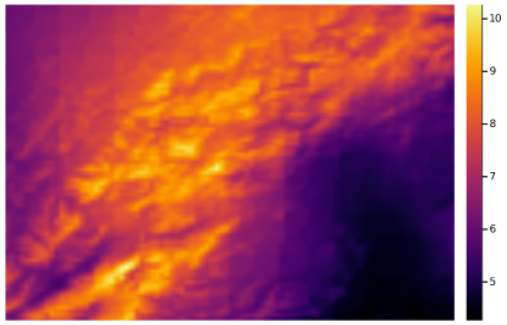
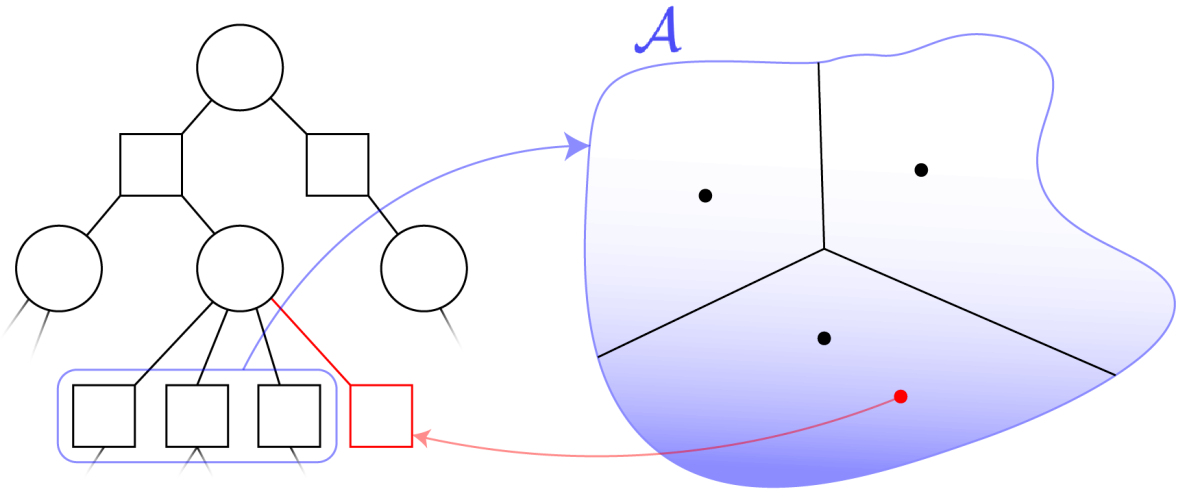


Figure 2: Wind Map. Figure shows wind map for Altamont Pass, CA at 100m altitude. The colors represent the average annual wind speed in m/s.

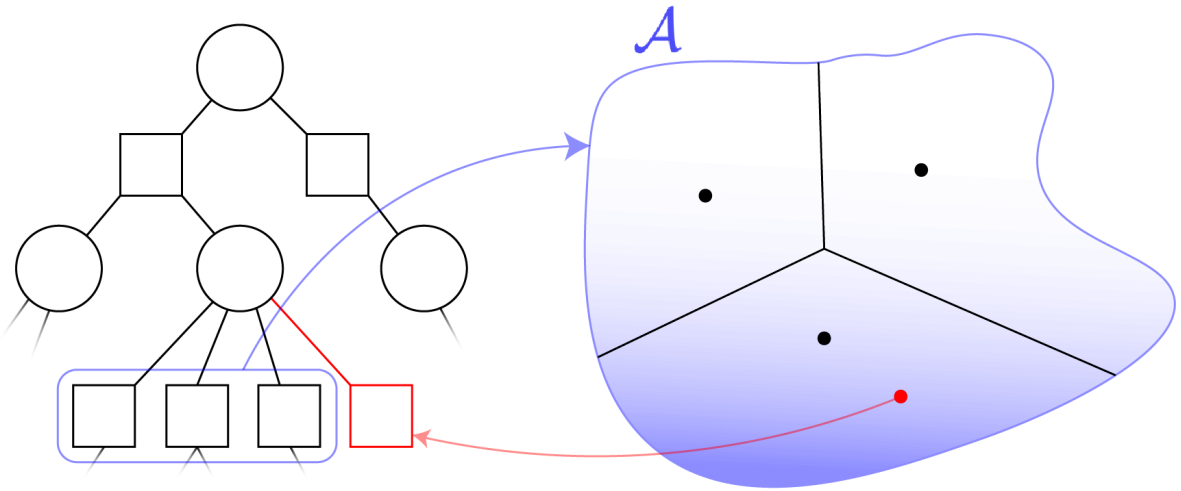
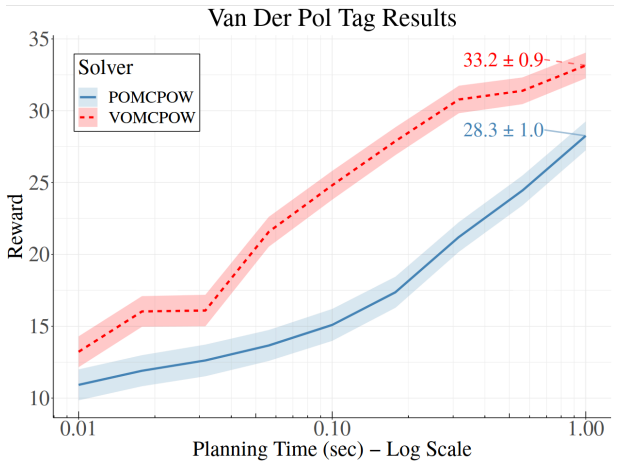
Algorithm	Queries	Score	Time (seconds)
POMCPOW	10	15708 ± 229	2.25 ± 0.07
	25	16234 ± 217	4.80 ± 0.07
	50	16374 ± 212	6.27 ± 0.08
	100	16018 ± 262	11.98 ± 0.07
	200	15787 ± 233	20.67 ± 0.09
BOMCP	10	18095 ± 183	2.55 ± 0.08
	25	18154 ± 158	5.21 ± 0.07
	50	18015 ± 163	6.71 ± 0.06
	100	18225 ± 119	13.39 ± 0.07
	200	18113 ± 157	25.14 ± 0.08
Expert	–	8130 ± 51	–

Voronoi Progressive Widening



Online Tree Search Planner Voronoi Progressive Widening

Voronoi Progressive Widening



Online Tree Search Planner Voronoi Progressive Widening